

# Data Communication & Computer Networks (CS24305) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

EXECUTIVE MASTER STUDY GUIDE & PROTOCOL BANK

## Data Communication & Computer Networks (CS24305)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024–25 Scheme)

### Complete Course Structure & Protocol Matrix

| Module     | Layer & Focus                | Key Formulations, RFCs & Protocols                                                                                |
|------------|------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Module I   | Physical Layer & Signals     | Nyquist & Shannon Capacity, Transmission Impairments, Line Coding (Manchester, AMI), Multiplexing (FDM, TDM, WDM) |
| Module II  | Data Link Layer & MAC        | Framing, CRC-32 Polynomial Derivations, Sliding Window ARQs (Go-Back-N, Selective Repeat), CSMA/CD & CSMA/CA      |
| Module III | Network Layer & Routing      | IPv4/IPv6 Header Analysis, CIDR Subnetting Calculations, Dijkstra Shortest Path, Bellman-Ford, OSPF, BGP-4        |
| Module IV  | Transport Layer Protocols    | TCP 3-Way Handshake, TCP State Machine, AIMD Congestion Control, Slow Start, Fast Retransmit, UDP Sockets         |
| Module V   | Application Layer & Security | DNS Hierarchical Resolution, HTTP/1.1 vs HTTP/2/3, TLS 1.3 Handshake, RSA Asymmetric Encryption, AES              |

### Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 5 modules with formal mathematical channel capacity proofs, step-by-step worked subnetting & CRC numericals, packet header diagrams, and model answers to BIT Mesra end-semester examination questions.

## Module I: Transmission Fundamentals & Physical Layer Architectures

1. Data Communication Components & Network Topologies (Mesh, Star, Bus, Ring)
2. The OSI 7-Layer Reference Model vs. TCP/IP 5-Layer Protocol Architecture
3. Analog vs. Digital Signals, Composite Signals & Fourier Spectral Analysis
4. Transmission Impairments (Attenuation, Distortion, Thermal & Cross-Talk Noise)
5. Nyquist Maximum Bit Rate Formula for Noiseless Channels & Multi-Level Signaling
6. Shannon Channel Capacity Theorem for Noisy Channels & Decibel (dB) Math
7. Line Coding Schemes: NRZ-L, NRZ-I, Manchester, Differential Manchester & AMI
8. Scrambling Techniques for Synchronization: B8ZS (North America) & HDB3 (Europe)
9. Transmission Media: Guided (Twisted Pair, Coax, Fiber) vs. Unguided (Radio, Micro)
10. Optical Fiber Physics: Step-Index, Graded-Index, Single-Mode vs. Multi-Mode
11. Multiplexing Fundamentals: FDM, Synchronous TDM, Statistical TDM & WDM
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

## Topic 1 & 2: Network Topologies & Layered Architectures

A **Computer Network** is an interconnected collection of autonomous computing devices capable of exchanging data frames and packets through transmission channels.

| OSI Layer       | Protocol Data Unit (PDU)       | Core Responsibilities & Header Additions                                                        | Standard Protocols & Hardware                    |
|-----------------|--------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 7. Application  | Message / Data                 | Network virtual terminal, file transfer, directory services, user interface.                    | HTTP, HTTPS, DNS, SMTP, FTP, SSH                 |
| 6. Presentation | Data                           | Data encryption/decryption (TLS), data compression (gzip), syntax translation (ASCII, UTF-8).   | TLS 1.3, JPEG, MPEG, ASN.1                       |
| 5. Session      | Data                           | Dialog control, token management, session synchronization checkpoints.                          | RPC, NetBIOS, PPTP, Sockets                      |
| 4. Transport    | Segment (TCP) / Datagram (UDP) | End-to-end process-to-process delivery, port addressing, flow control, segment reassembly.      | TCP, UDP, SCTP, QUIC                             |
| 3. Network      | Packet / Datagram              | Host-to-host logical addressing (IPv4/IPv6), packet routing, fragmentation, congestion control. | IP, ICMP, OSPF, BGP (Routers)                    |
| 2. Data Link    | Frame                          | Hop-to-hop node delivery, physical MAC addressing, framing, CRC error detection, MAC access.    | Ethernet (802.3), Wi-Fi (802.11), PPP (Switches) |
| 1. Physical     | Raw Bits (0/1)                 | Transmission of uninterpreted bit streams over physical media, signal encoding, baud rates.     | Manchester, NRZ, RS-232, RJ-45 (Hubs, Cables)    |

## Topic 4 – 6: Channel Capacity Theorems & Decibel Formulations

### 1. Nyquist Bit Rate Formula (For Noiseless Channels):

$$\text{BitRate}_{\max} = 2 \times B \times \log_2(M) \quad \text{bps}$$

Where  $B$  is the analog bandwidth in Hertz (Hz), and  $M$  is the number of distinct signal voltage levels.

### 2. Shannon Channel Capacity Theorem (For Noisy Gaussian Channels):

$$C = B \times \log_2(1 + \text{SNR}) \quad \text{bps}$$

Where  $B$  is the bandwidth (Hz), and  $\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$  is the linear Signal-to-Noise Ratio.

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \implies \text{SNR} = 10^{\frac{\text{SNR}_{\text{dB}}}{10}}$$



### Step-by-Step Solved Problem: Shannon & Nyquist Capacity Calculation

**Problem:** A telephone line has a bandwidth of 3000 Hz and an  $\text{SNR}_{\text{dB}} = 30$  dB.

#### 1. Step 1: Compute Linear SNR from dB:

$$\text{SNR} = 10^{\frac{30}{10}} = 10^3 = \mathbf{1000}$$

#### 2. Step 2: Apply Shannon's Capacity Formula:

$$C = 3000 \times \log_2(1 + 1000) = 3000 \times \log_2(1001) \approx 3000 \times 9.967 = \mathbf{29,901 \text{ bps}} (\approx 30 \text{ kbps})$$

#### 3. Step 3: Find Number of Signal Levels $M$ Required by Nyquist to achieve $C$ :

$$29901 = 2 \times 3000 \times \log_2(M) \implies \log_2(M) = \frac{29901}{6000} \approx 4.9835 \implies M = 2^{4.9835} \approx \mathbf{32 \text{ levels}}$$

## Topic 7 & 8: Digital Signal Line Coding Schemes

| Line Coding Scheme                | Voltage Transition Rule                                                                                                | Key Advantages & Tradeoffs                                                                               |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1. NRZ-L (Level)                  | Bit '0' $\rightarrow$ Positive Voltage ( $+V$ ), Bit '1' $\rightarrow$ Negative Voltage ( $-V$ ).                      | Simple; suffers from DC component and baseline wander on long strings of '0's or '1's.                   |
| 2. NRZ-I (Invert)                 | Bit '1' $\rightarrow$ Transition at start of bit interval; Bit '0' $\rightarrow$ No transition.                        | Self-clocking for '1's; still loses clock synchronization on continuous strings of '0's.                 |
| 3. Manchester (IEEE 802.3)        | Bit '0' $\rightarrow$ High-to-Low transition at bit center; Bit '1' $\rightarrow$ Low-to-High transition at center.    | Zero DC component, 100% self-clocking; requires $2\times$ bandwidth ( $B = 2 \times \text{data rate}$ ). |
| 4. Differential Manchester        | Always transitions at center. Bit '0' $\rightarrow$ Transition at start; Bit '1' $\rightarrow$ No transition at start. | Highly immune to wire polarity reversal; used in Token Ring LANs (IEEE 802.5).                           |
| 5. AMI (Alternate Mark Inversion) | Bit '0' $\rightarrow$ Zero Voltage ( $0V$ ); Bit '1' $\rightarrow$ Alternating $+V$ and $-V$ .                         | Zero DC component; long runs of '0's lose synchronization (resolved by B8ZS/HDB3 scrambling).            |



### Q1. State the difference between OSI and TCP/IP reference models across 5 engineering parameters. (8 Marks)

1. **Layer Count:** OSI has 7 layers (Session & Presentation separate); TCP/IP has 4 or 5 layers (combines Session, Presentation, Application).
2. **Model Origins:** OSI is a theoretical conceptual standard developed by ISO before protocols were written; TCP/IP was designed pragmatically alongside the ARPANET implementation.
3. **Service Orientation:** OSI strictly separates Service, Interface, and Protocol; TCP/IP merges services into protocols directly.
4. **Network Layer Communication:** OSI supports both Connection-Oriented and Connectionless at Network layer; TCP/IP supports only Connectionless (IP) at Network layer.
5. **Transport Layer Communication:** OSI supports only Connection-Oriented at Transport; TCP/IP supports both Connection-Oriented (TCP) and Connectionless (UDP).

## Module II: Data Link Layer, Error Control & Flow Control ARQs

1. Data Link Layer Design Issues & Service Models to Network Layer
2. Framing Mechanisms: Byte Count, Byte Stuffing & Bit Stuffing with Flag `01111110`
3. Error Detection: Simple Parity, 2D Parity & Internet Checksum Algorithm
4. Cyclic Redundancy Check (CRC-32): Polynomial Division & Full Worked Traces
5. Error Correction: Hamming Code  $(n, k)$  Derivations & Parity Check Matrices
6. Flow Control Mechanics: Stop-and-Wait Protocol & Channel Efficiency  $(\eta)$
7. Sliding Window Concepts: Bandwidth-Delay Product & Piggybacking
8. Go-Back-N (GBN) ARQ Protocol: Sender/Receiver Windows  $(W_s \leq 2^m - 1)$
9. Selective Repeat (SR) ARQ Protocol: Independent ACKs & Windows  $(W_s \leq 2^{m-1})$
10. Channel Utilization & Efficiency Proofs:  $\eta = \frac{W}{1+2a}$  where  $a = \frac{T_p}{T_t}$
11. High-Level Data Link Control (HDLC) Frame Formats & PPP Architecture
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

### Topic 1 & 2: Data Link Framing Mechanisms

The Data Link Layer converts the raw bit stream delivered by the Physical Layer into discrete, manageable packets called **Frames**:

| Framing Method                        | Delimiting Mechanism & Bit/Byte Manipulation                                                                                                                        | Key Failure Modes & Mitigations                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1. Character / Byte Count             | Header field contains an integer specifying total number of bytes in frame.                                                                                         | Fatal synchronization loss: if a transmission error alters the count byte, all subsequent frames are misframed. |
| 2. Byte Stuffing (Character-Oriented) | Frames bounded by `FLAG` bytes (`0x7E`). Any occurrence of `FLAG` or escape byte `ESC` (`0x7D`) in payload is prefixed with an `ESC` byte.                          | Tied to specific 8-bit character encodings (ASCII).                                                             |
| 3. Bit Stuffing (Bit-Oriented — HDLC) | Frames delimited by flag pattern `01111110` (6 consecutive `1`s). Whenever sender observes 5 consecutive `1`s in data payload, it unconditionally stuffs a `0` bit. | Completely code-transparent; handles arbitrary arbitrary binary streams.                                        |

### Topic 4: Cyclic Redundancy Check (CRC) Polynomial Division

**CRC Transmission Invariants:** Let data bit sequence be represented by polynomial  $D(x)$  of length  $k$ , and Generator Polynomial be  $G(x)$  of degree  $r$ . 1. Append  $r$  zero bits to  $D(x) \implies D(x) \cdot x^r$ . 2. Divide  $D(x) \cdot x^r$  by  $G(x)$  using modulo-2 arithmetic (XOR binary division). 3. The remainder  $R(x)$  of length  $r$  is the CRC Checksum. Transmitted codeword  $T(x) = D(x) \cdot x^r \oplus R(x)$ .

### Step-by-Step Solved Problem: CRC-4 Calculation on Data `1101011011` with Generator $G(x) = x^4 + x + 1$

**Step 1: Convert Generator to Binary String:**  $G(x) = 1 \cdot x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x^1 + 1 \cdot x^0 \implies 10011$  (Degree  $r = 4$ ).

**Step 2: Append  $r = 4$  zeros to Data ( $D = 1101011011$ ):** Augmented Dividend: **11010110110000**.

**Step 3: Modulo-2 Polynomial XOR Division:**

```

      1100001010 (Quotient)
10011 ) 11010110110000
      10011
      ----
      10011
      10011
      ----
      000010110
        10011
        ----
        010100
          10011
          ----
          01110 ← Remainder R = 1110
  
```

**Step 4: Transmitted Codeword:**  $11010110110000 \oplus 1110 = 11010110111110$ .

## Topic 6 – 10: Sliding Window Flow Control ARQ Protocols

| Protocol                 | Sender Window ( $W_s$ ) | Receiver Window ( $W_r$ )         | Channel Efficiency ( $\eta$ )                               |
|--------------------------|-------------------------|-----------------------------------|-------------------------------------------------------------|
| 1. Stop-and-Wait         | $W_s = 1$               | $W_r = 1$                         | $\eta = \frac{T_t}{T_t + 2T_p} = \frac{1}{1 + 2a}$          |
| 2. Go-Back-N (GBN)       | $W_s = 2^m - 1$         | $W_r = 1$ (Cumulative ACK)        | $\eta = \frac{W_s}{1 + 2a}$ (if $W_s < 1 + 2a$ , else 100%) |
| 3. Selective Repeat (SR) | $W_s = 2^{m-1}$         | $W_r = 2^{m-1}$ (Independent ACK) | $\eta = \frac{W_s}{1 + 2a}$                                 |

Where  $m$  = number of sequence number bits,  $T_t = \frac{L}{B}$  = transmission time,  $T_p = \frac{d}{v}$  = propagation delay,  $a = \frac{T_p}{T_t}$ .

**Q1. Why must the maximum sender window size in Selective Repeat ARQ satisfy  $W_s \leq 2^{m-1}$ ? (8 Marks)**

If  $W_s + W_r > 2^m$ , an ambiguous overlap occurs between the old sequence number window and the new sequence number window.

*Proof:* Suppose  $m = 2$  (sequence numbers 0, 1, 2, 3) and  $W_s = W_r = 3$ . Sender transmits frames 0, 1, 2. Receiver receives all 3 and shifts window to [3, 0, 1], sending ACKs 0, 1, 2. If all ACKs are lost in transit, sender times out and retransmits frame 0. The receiver, expecting new frame 0 in its current window [3, 0, 1], will accept the duplicate old frame 0 as new data, causing undetected duplicate data corruption! To prevent window overlap,  $W_s + W_r \leq 2^m \implies W_s \leq 2^{m-1}$ .

## Module III: Medium Access Control (MAC), Local Area Networks & Wireless LANs

1. Channel Allocation Problem: Static FDM/TDM vs. Dynamic Random Access
2. Pure ALOHA Protocol Mathematical Throughput Derivation ( $S = Ge^{-2G} \Rightarrow 18.4\%$ )
3. Slotted ALOHA Protocol Mathematical Throughput Derivation ( $S = Ge^{-G} \Rightarrow 36.8\%$ )
4. Carrier Sense Multiple Access (CSMA): 1-Persistent, Non-Persistent,  $p$ -Persistent
5. CSMA with Collision Detection (CSMA/CD – IEEE 802.3 Ethernet Standard)
6. Minimum Frame Size Derivation:  $L_{\min} = 2 \times R \times \frac{d}{v}$  in Ethernet
7. Truncated Binary Exponential Backoff Algorithm ( $0 \leq k < 2^i$ )
8. Standard Ethernet IEEE 802.3 Frame Structure & Physical MAC 48-Bit Addresses
9. Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps) & 10-Gigabit Standards
10. Wireless LANs (IEEE 802.11 Wi-Fi Architecture & CSMA/CA with RTS/CTS)
11. The Hidden Station & Exposed Station Problems with Virtual Carrier Sensing (NAV)
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

### Topic 2 & 3: ALOHA Protocols & Poisson Throughput Proofs

**1. Pure ALOHA Throughput Derivation:** Let  $G$  be the total offered traffic load (frames per frame transmission time  $T$ ). - Vulnerable Period in Pure ALOHA =  $2 \times T$  (any frame generated in  $(t - T, t + T)$  collides). - Under Poisson distribution, probability of zero other frame arrivals is  $P(0) = e^{-2G}$ . - Throughput:

$$S = G \cdot e^{-2G}$$

- Maximum Throughput occurs at  $\frac{dS}{dG} = e^{-2G}(1 - 2G) = 0 \Rightarrow G = 0.5$ :

$$S_{\max} = 0.5 \cdot e^{-1} = \frac{1}{2e} \approx 0.184 \quad (18.4\%)$$

**2. Slotted ALOHA Throughput Derivation:** - Frames can only be transmitted at discrete slot boundaries. - Vulnerable Period =  $1 \times T$ . - Throughput:

$$S = G \cdot e^{-G}$$

- Maximum Throughput occurs at  $G = 1.0$ :

$$S_{\max} = 1.0 \cdot e^{-1} = \frac{1}{e} \approx 0.368 \quad (36.8\%)$$

### Topic 5 & 6: CSMA/CD & Minimum Frame Size Equation

In **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)**, a transmitting station continuously listens to the physical cable while transmitting.

#### The Fundamental CSMA/CD Minimum Frame Size Rule

To guarantee that a station detects a collision before it finishes transmitting its frame, the transmission time  $T_t$  must be at least twice the maximum round-trip propagation delay  $2 \times T_p$ :

$$T_t \geq 2 \times T_p \Rightarrow \frac{L_{\min}}{B} \geq 2 \times \frac{d}{v} \Rightarrow L_{\min} = 2 \times B \times \frac{d}{v}$$

Where  $L_{\min}$  is the minimum frame size in bits,  $B$  is the bandwidth in bps,  $d$  is the maximum network length in meters, and  $v$  is the signal propagation velocity in the cable ( $\approx 2 \times 10^8$  m/s).



### Step-by-Step Solved Problem: Classical 10 Mbps Ethernet 64-Byte Minimum Frame Size

For standard 10 Mbps Ethernet with maximum collision domain distance 2500 meters and 4 repeaters:

$$2T_p \approx 51.2 \mu s$$

$$L_{\min} = 10 \times 10^6 \text{ bps} \times 51.2 \times 10^{-6} \text{ s} = 512 \text{ bits} = \mathbf{64 \text{ bytes}}$$

Frames smaller than 64 bytes are padded with zeroes up to 64 bytes (Runt Frames < 64 bytes are discarded as collisions).

## Topic 10 & 11: Wireless LANs (IEEE 802.11) & The Hidden Station Problem

| Wireless Phenomenon        | Geometric Problem Topology                                                                                                                                                                                                             | CSMA/CA RTS/CTS Solution                                                                                                                                                                                 |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Hidden Station Problem  | Station $A$ and Station $C$ can both transmit to Base Station $B$ , but $A$ and $C$ are out of range of each other. If both sense carrier simultaneously, they detect an idle channel and transmit, causing severe collisions at $B$ . | <b>RTS/CTS Handshake:</b> $A$ sends `RTS` (Request to Send). $B$ broadcasts `CTS` (Clear to Send). $C$ overhears $B$ 's `CTS` and sets its <b>Network Allocation Vector (NAV)</b> to defer transmission. |
| 2. Exposed Station Problem | Station $B$ is transmitting to $A$ . Station $C$ wants to transmit to $D$ . $C$ senses the medium, detects $B$ 's transmission, and wrongly defers, even though $C \rightarrow D$ would not interfere with $B \rightarrow A$ .         | Resolved by spatial channel reuse and directional antennas.                                                                                                                                              |

## Top BIT Mesra Exam Questions & Answers (Module III)

### Q1. Explain the Truncated Binary Exponential Backoff Algorithm used in Ethernet. (8 Marks)

After  $i$  collisions for a frame:

1. The station chooses a random integer  $k$  from the uniform range  $0 \leq k < 2^c$ , where  $c = \min(i, 10)$ .
2. The station waits  $k \times \text{SlotTime}$  ( $k \times 51.2 \mu s$  in 10 Mbps Ethernet) before attempting retransmission.
3. After 10 collisions, the backoff interval freezes at  $2^{10} - 1 = 1023$  slots.
4. If the frame experiences 16 consecutive collisions ( $i = 16$ ), the transmission is aborted and an error is reported to the upper layer.

## Module IV: Network Layer, IP Addressing & Routing Algorithms

1. Network Layer Service Models: Virtual-Circuit vs. Datagram Packet Switching
2. IPv4 Header Format: 20-Byte Field Breakdown & Fragmentation Fields (Offset, Flags)
3. Classful IP Addressing (Classes A, B, C, D, E & Special Loopback / Private Blocks)
4. Subnetting & Classless Inter-Domain Routing (CIDR) Slash Notation ( $/n$ )
5. Longest Prefix Match Algorithm in Forwarding Tables with Worked Lookups
6. IPv6 Protocol Architecture: 40-Byte Fixed Base Header & Extension Headers
7. IPv4 to IPv6 Transition Mechanisms: Dual-Stack, Tunneling & Header Translation
8. Unicast Routing Principles: Intra-AS (IGP) vs. Inter-AS (EGP) Autonomous Systems
9. Distance Vector Routing (DVR): Bellman-Ford Equation & Count-to-Infinity Problem
10. Solutions to Count-to-Infinity: Split Horizon & Poisoned Reverse Mechanisms
11. Link State Routing (LSR): Link State Advertisements & Dijkstra's Algorithm Tree
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

### Topic 2 – 4: IPv4 Datagram Format & CIDR Subnetting

| IPv4 Header Field             | Bit Width        | Functional Purpose & Transmission Invariant                                                                                          |
|-------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Version                       | 4 bits           | Identifies IP version (`0100` for IPv4).                                                                                             |
| Header Length (IHL)           | 4 bits           | Length of header in 32-bit (4-byte) words. Minimum value is 5 (20 bytes).                                                            |
| Total Length                  | 16 bits          | Entire datagram length (Header + Data) in bytes. Max size = 65,535 bytes.                                                            |
| Identification, Flags, Offset | 16 + 3 + 13 bits | Handles MTU packet fragmentation. Fragment Offset is measured in 8-byte units. `DF` = Don't Fragment, `MF` = More Fragments.         |
| Time to Live (TTL)            | 8 bits           | Hop counter decremented by 1 at each router to prevent infinite forwarding loops. Discarded with ICMP Time Exceeded when $TTL = 0$ . |
| Protocol                      | 8 bits           | Demultiplexing to transport layer (`6` for TCP, `17` for UDP, `1` for ICMP).                                                         |
| Header Checksum               | 16 bits          | 1's complement sum of header fields recalculated at every router as TTL decrements.                                                  |

### Step-by-Step Solved Problem: CIDR Subnet Allocation

**Problem:** An organization is assigned the network block `200.10.20.0/24`. It needs to divide this block into 4 equal subnets for 4 departments.

#### 1. Step 1: Determine Additional Subnet Bits ( $k$ ):

$$2^k \geq 4 \implies k = 2 \text{ bits}$$

New Subnet Mask:  $/24 + 2 = /26 \implies \mathbf{255.255.255.192}$ .

#### 2. Step 2: Compute Total and Usable Host Addresses per Subnet:

$$\text{Total Addresses per Subnet} = 2^{32-26} = 2^6 = \mathbf{64}$$

$$\text{Usable Host Addresses} = 64 - 2 = \mathbf{62} \quad (\text{subtracting Network ID and Directed Broadcast})$$

#### 3. Step 3: Subnet Allocation Table:

| Subnet | Network Address    | Usable Host IP Range              | Broadcast Address |
|--------|--------------------|-----------------------------------|-------------------|
| Dept 1 | `200.10.20.0/26`   | `200.10.20.1` - `200.10.20.62`    | `200.10.20.63`    |
| Dept 2 | `200.10.20.64/26`  | `200.10.20.65` - `200.10.20.126`  | `200.10.20.127`   |
| Dept 3 | `200.10.20.128/26` | `200.10.20.129` - `200.10.20.190` | `200.10.20.191`   |
| Dept 4 | `200.10.20.192/26` | `200.10.20.193` - `200.10.20.254` | `200.10.20.255`   |

## Topic 9 – 11: Routing Algorithms (Distance Vector vs. Link State)

| Feature                 | Distance Vector Routing (DVR — RIP)                                     | Link State Routing (LSR — OSPF)                                                          |
|-------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Mathematical Algorithm  | <b>Bellman-Ford Equation:</b><br>$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$ | <b>Dijkstra's Shortest Path Algorithm:</b> Computes full shortest-path tree from source. |
| Information Exchanged   | Vector of estimated distances to all destinations.                      | Exact state and cost of directly attached links (LSA).                                   |
| Destination of Updates  | Sent only to directly adjacent neighbors.                               | Flooded globally to all routers in the entire network area.                              |
| Convergence & Stability | Slow convergence; prone to <b>Count-to-Infinity</b> routing loops.      | Fast convergence; completely immune to routing loops.                                    |

**Q1. Explain the Count-to-Infinity problem in Distance Vector Routing and how Split Horizon with Poison Reverse resolves it. (8 Marks)**

- **Count-to-Infinity Problem:** When a link  $A - B$  goes down, router  $A$  sets its cost to  $\infty$ . However, neighbor  $C$  may still advertise that it has a path to  $A$  via  $B$  with cost 2. Router  $B$  receives this and updates its cost to  $A$  as  $2 + 1 = 3$ , advertising it back to  $C$ .  $C$  updates to  $3 + 1 = 4$ , and both routers slowly increment their distance vectors step-by-step up to  $\infty$  (16 in RIP).
- **Split Horizon with Poison Reverse:** If router  $C$  routes traffic to destination  $X$  through neighbor  $B$ ,  $C$  advertises its distance to  $X$  back to  $B$  as  $\infty$  (Poisoned Reverse). This prevents  $B$  from wrongly routing traffic back through  $C$  during link failures.

## Module V: Transport Layer, Application Services & Network Security

1. Transport Layer Port Multiplexing & Sockets Interface (TCP vs. UDP)
2. TCP Segment Anatomy: Sequence Numbers, Acknowledgments & Control Flags
3. TCP 3-Way Handshake Connection Establishment & 4-Way FIN Teardown
4. TCP Flow Control (Receiver Window `rwnd` & Nagle's Algorithm)
5. TCP Congestion Control: Slow Start, Congestion Avoidance (AIMD), Fast Retransmit
6. TCP Fast Recovery & Cubic / BBR Modern Congestion Mechanisms
7. Domain Name System (DNS): Hierarchical Namespace, Root Servers & Records (A, AAAA, MX)
8. Hypertext Transfer Protocol Evolution: HTTP/1.1 vs. HTTP/2 vs. HTTP/3 (QUIC)
9. Email Architecture: SMTP, POP3, IMAP & MIME Extensions
10. Cryptographic Fundamentals: Symmetric (AES) vs. Asymmetric (RSA) Cryptography
11. RSA Public-Key Algorithm Mathematical Derivation & Worked Encryption Traces
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

### Topic 1 – 3: TCP Protocol Architecture & 3-Way Handshake

| TCP Control Flag | Bit Position   | Operational Role in Connection State Machine                                                               |
|------------------|----------------|------------------------------------------------------------------------------------------------------------|
| SYN              | Synchronize    | Initiates a connection; synchronizes initial sequence numbers (ISN). Consumes 1 sequence number.           |
| ACK              | Acknowledgment | Validates the 32-bit Acknowledgment Number field indicating next expected byte.                            |
| FIN              | Finish         | Sender has finished sending data and requests connection termination. Consumes 1 sequence number.          |
| RST              | Reset          | Abruptly terminates an invalid or unauthorized connection request.                                         |
| PSH              | Push           | Instructs receiver to deliver buffered data to application immediately without waiting for buffer to fill. |
| URG              | Urgent         | Validates the Urgent Pointer field for out-of-band high-priority data (e.g., `Ctrl+C`).                    |

#### Step-by-Step Solved Problem: TCP 3-Way Handshake Connection Sequence

1. **Client → Server (SYN)**: Client chooses random initial sequence number  $x$ :

Segment 1: [**SYN** = 1, **Seq** =  $x$ ]

2. **Server → Client (SYN-ACK)**: Server acknowledges client's ISN ( $x + 1$ ) and chooses its own ISN  $y$ :

Segment 2: [**SYN** = 1, **ACK** = 1, **Seq** =  $y$ , **Ack** =  $x + 1$ ]

3. **Client → Server (ACK)**: Client acknowledges server's ISN ( $y + 1$ ):

Segment 3: [**ACK** = 1, **Seq** =  $x + 1$ , **Ack** =  $y + 1$ ]

*Connection established in ESTABLISHED state.*

## Topic 5 & 6: TCP Congestion Control Mechanisms (AIMD)

### The 4 Standard Phases of TCP Congestion Control (Tahoe / Reno):

1. **Slow Start:** Initial congestion window  $\text{cwnd} = 1 \text{ MSS}$ . Doubles every Round-Trip Time ( $\text{cwnd} \leftarrow \text{cwnd} \times 2$ ) exponentially until  $\text{cwnd} \geq \text{ssthresh}$  (Slow Start Threshold).
2. **Congestion Avoidance (Additive Increase):** Increases linearly by 1 MSS per RTT ( $\text{cwnd} \leftarrow \text{cwnd} + 1$ ).
3. **Triple Duplicate ACK Event (Fast Retransmit & Recovery):** -  $\text{ssthresh} \leftarrow \frac{\text{cwnd}}{2}$  -  $\text{cwnd} \leftarrow \text{ssthresh} + 3 \text{ MSS}$  (TCP Reno skips Slow Start and continues linearly).
4. **Timeout Event (Severe Congestion):** -  $\text{ssthresh} \leftarrow \frac{\text{cwnd}}{2}$  -  $\text{cwnd} \leftarrow 1 \text{ MSS}$  (Drops back to Slow Start).

## Topic 10 & 11: Network Security & RSA Public-Key Cryptography

### RSA Algorithm Mathematical Formulations (Rivest, Shamir, Adleman, 1977):

1. Select two distinct large prime numbers  $p$  and  $q$ .
2. Compute modulus  $n = p \times q$ .
3. Compute Euler's Totient  $\phi(n) = (p - 1)(q - 1)$ .
4. Choose public exponent  $e$  such that  $1 < e < \phi(n)$  and  $\text{gcd}(e, \phi(n)) = 1$ .
5. Compute private decryption exponent  $d$  using Extended Euclidean Algorithm:

$$d \equiv e^{-1} \pmod{\phi(n)} \implies (d \times e) \pmod{\phi(n)} = 1$$

6. **Public Key:**  $KU = \{e, n\}$ , **Private Key:**  $KR = \{d, n\}$ .
7. **Encryption:** Ciphertext  $C = M^e \pmod{n}$ .
8. **Decryption:** Plaintext  $M = C^d \pmod{n}$ .

### Step-by-Step Solved Problem: Complete RSA Encryption & Decryption Trace

Let primes  $p = 3$ ,  $q = 11$ , and Plaintext message  $M = 7$ .

1.  $n = 3 \times 11 = 33$ .
2.  $\phi(n) = (3 - 1)(11 - 1) = 2 \times 10 = 20$ .
3. Choose  $e = 7$  ( $\text{gcd}(7, 20) = 1$ ).
4. Find  $d$ :  $(d \times 7) \pmod{20} = 1 \implies 7d = 21 \implies d = 3$ .
5. **Encryption:**  $C = 7^7 \pmod{33} = 823543 \pmod{33} = 4$ .
6. **Decryption:**  $M = 4^3 \pmod{33} = 64 \pmod{33} = 7$ . (Original plaintext recovered!).

## Top BIT Mesra Exam Questions & Answers (Module V)

### Q1. Compare HTTP/1.1, HTTP/2, and HTTP/3 across 4 architectural parameters. (8 Marks)

1. **Transport Layer:** HTTP/1.1 and HTTP/2 run over standard TCP; HTTP/3 runs over QUIC (UDP-based).
2. **Multiplexing:** HTTP/1.1 uses sequential request-response pipelining (suffers from Head-of-Line blocking); HTTP/2 uses binary frame multiplexing over a single TCP connection; HTTP/3 achieves stream-level multiplexing without TCP Head-of-Line blocking.
3. **Header Compression:** HTTP/1.1 uses plain text headers without compression; HTTP/2 uses HPACK; HTTP/3 uses QPACK.
4. **Handshake Latency:** HTTP/1.1 & HTTP/2 require 2-3 RTTs (TCP + TLS); HTTP/3 achieves 0-RTT connection resumption.